

Management Information Systems

12. Designing Databases and Human-Computer Interaction

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Designing Databases and Human-Computer Interaction

- Designing Databases (Chapter 13, Textbook)
 - Relational Databases
 - Normalization
 - Guidelines for Master File/Database Relation Design
 - Retrieving and Presenting Database Data
 - Data Warehouses and Business Intelligence
- Human-Computer Interaction and UX Design (Chapter 14, Textbook)
 - Understanding human–computer interaction (HCI)
 - Designing Useful User Interfaces
 - Designing Onscreen Dialog for HCI
 - Understanding the Importance of User Feedback
 - Designing Ecommerce Websites.
 - Formulating Queries for Searching the Web



Where We Are

- Re-thinking the Big Picture of this Course
 - Lectures 1~6: Understanding the problem (people, processes, organization, requirements)
 - Lectures 7~10: Analyzing the problem and modeling the solution (DFDs, UML, prototyping)
 - Lecture 11: Output & Input - what users see, what users enter
 - **Lecture 12: Data & Interaction** - what the system **remembers** and how users **talk to it**
 - Lecture 13: Quality, security, implementation



Our Roadmap

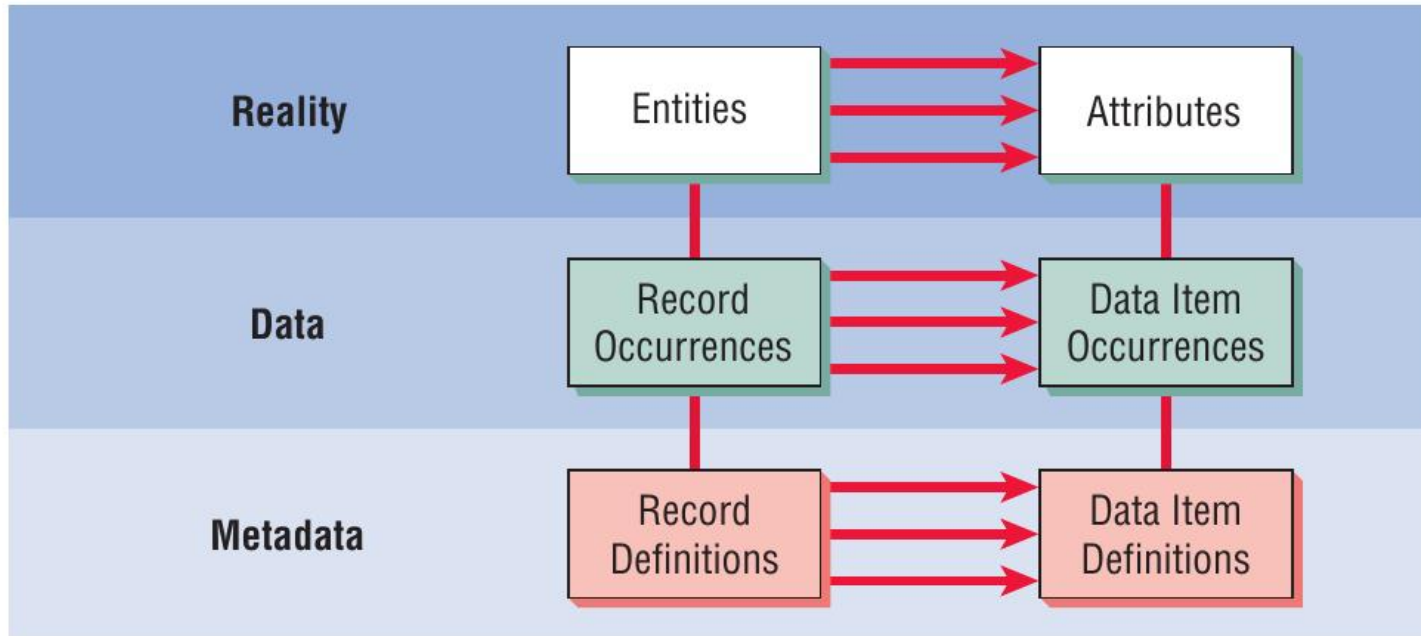
- Databases: the Memory of the System
 - Relational Databases: from reality to data
 - Normalization (规范化): ensure data consistency
 - OLTP vs. OLAP: consistency vs. analyzability
 - NoSQL and Vector Databases: towards more natural interaction with users
- HCI: the Mouth and Ears of the System
 - HCI as Fit: the core principle
 - Types of User Interfaces
 - Types of Feedback
 - Agentic Interface
- An Integrated Case
 - The Catalyst Project (启航杯)



Databases: the Memory of the System



Reality, Data, and Metadata

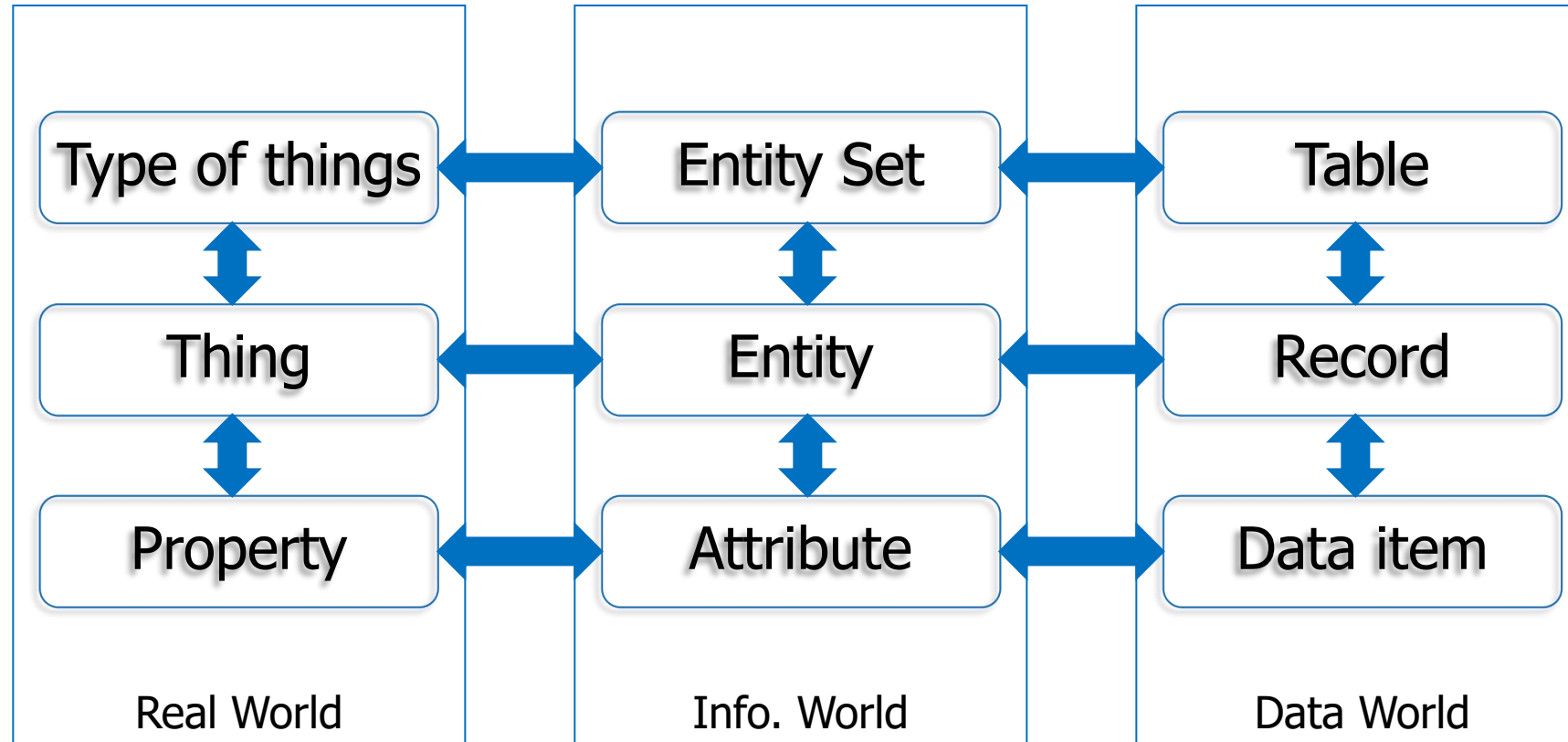


A student named 张三,
born 2003

Row: (20230001, "张三",
"2003-05-12")

Table Student (
id INT,
name VARCHAR,
dob DATE)

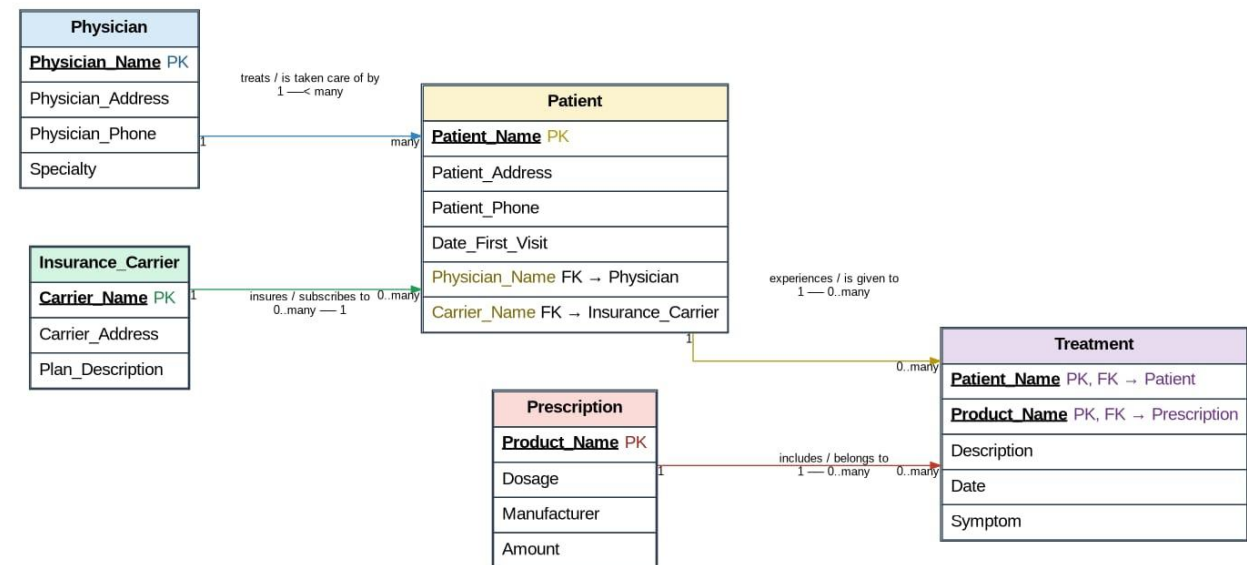
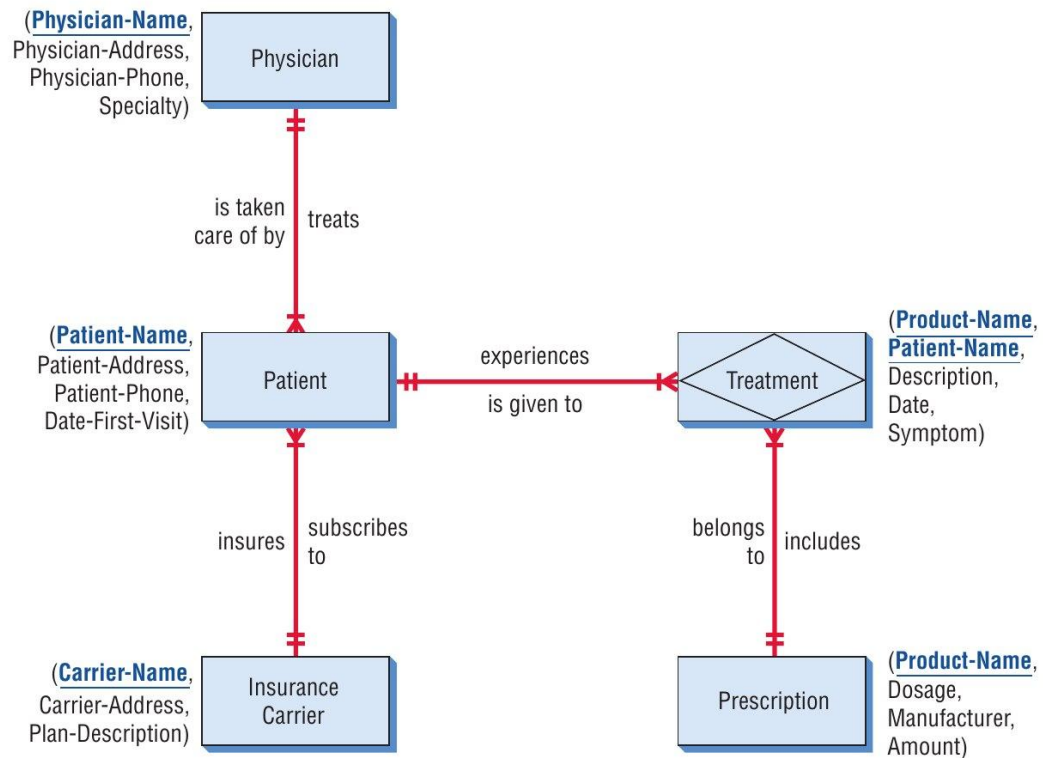
Relational Databases: from reality to data



E-R Diagram

Relational Schema

From E-R Diagram to Relational Schema



Relational Database

ITEM PRICE

ITEM-#	TITLE	PRICE
B235	Guys and Dolls	8.99
B521	My Fair Lady	6.99
B894	42nd Street	10.99
B992	A Chorus Line	10.99

ORDER

ORDER-#	LAST NAME	I	STREET ADDRESS	CITY	ST	CHARGE ACCT
10784	MacRae	G	2314 Curly Circle	Lincoln	NE	45-4654-76
10796	Jones	S	34 Dream Lane	Oklahoma City	OK	44-9876-74
11821	Preston	R	1008 Madison Ave.	River City	IA	34-7642-64
11845	Channing	C	454 Harmonia St.	New York	NY	34-0876-87
11872	Kiley	R	765 Dulcinea Drive	La Mancha	CA	65-8798-87

ITEM STATUS

ITEM-#	ORDER-#	STATUS
B235	10784	Shipped 5/12
B235	19796	Shipped 5/14
B235	11872	In Process
B521	11821	In Process
B894	11845	Backordered
B894	11872	Shipped 5/12
B992	10784	Shipped 5/12



Normalization (规范化)

- What's wrong with this database?

OrderID	Customer	CustPhone	Product	UnitPrice	Qty
1001	张三	138...	iPhone	6999	1
1002	张三	138...	AirPods	1299	2
1003	张三	139...	iPhone	7099	1

Three anomalies

- Update (phone only updated in one row)
- Insert (can't add customer without order)
- Delete (last order gone → customer gone)

Normalization = eliminating multiple copies of one truth.



Normal Forms

Form	Rule	Plain Explanation
1NF	Atomic values, no repeating groups	One cell, one fact
2NF	1NF + no partial dependency on composite key	Don't mix two things in one table
3NF	2NF + no transitive dependency	Non-key attributes depend on the key only

- Practice:
 - Aim for 3NF by default
 - Denormalize INTENTIONALLY when query performance demands it
 - Normalize for correctness
 - Denormalize for performance
 - Document both



OLTP vs. OLAP

	OLTP (Operational)	OLAP (Analytical)
Purpose	Run the business	Understand the business
Focus	Consistency	Analyzability
Workload	Many small transactions	Few large queries
Schema	Normalized (2NF - 3NF)	Denormalized
Example	Place an order	Which products grew 30% YoY in tier-3 cities?



NoSQL Databases

Type	Example	Sacrifices	Gains
Key-Value	Redis, DynamoDB	Query expressiveness	Brutal speed
Document	MongoDB	Strict schema	Flexible nested data
Column-Family	Cassandra, HBase	Joins, transactions	Massive write scale
Graph	Neo4j	Tabular thinking	Native relationship traversal



Key-value Databases (e.g., Redis)

- A key-value database stores data as a collection of pairs:
 - key $\rightarrow$ value

```
session:abc123  $\rightarrow$  {  
  "user_id": "U1001",  
  "name": "Alice Chen",  
  "role": "student"  
}  
.....  
session:xyz999  $\rightarrow$  {  
  "user_id": "U9999",  
  "name": "John Smith",  
  "role": "administrator"  
}
```

Query:
GET session:abc123

```
{  
  "user_id": "U1001",  
  "name": "Alice Chen",  
  "role": "student"  
}
```

Document Databases (e.g., MongoDB)

- A document database also stores data using an ID or key
 - but the value is a structured document
 - usually JSON-like

```
{
  "_id": "S1001",
  "student_name": "Alice Chen",
  "major": "Information Systems",
  "enrollments": [
    {
      "course_id": "DB101",
      "course_title": "Database Systems",
      "grade": "A"
    }
  ]
},
.....
```

Query:

```
db.students.find({
  major: "Information Systems"
});
```

Query:

```
db.students.find({
  "enrollments.course_id": "DB101"
});
```



Vector Databases (向量数据库)

- Core insight: Store MEANING as COORDINATES, not text.
 - Imagine a 2-D "meaning map":

"Apple iPhone"	(90, 20)	┐	almost identical!
"Huawei phone"	(88, 22)	┘	
"Apple juice"	(15, 80)	←	far away
 - "Apple iPhone" and "Apple juice" share a WORD but live FAR APART.
 - "Apple iPhone" and "Huawei phone" share NO WORDS but live RIGHT NEXT to each other.
- A neural network (the "embedding model") learns these coordinates by reading the entire Internet.
- Traditional DB: find books by ISBN NUMBER
- Vector DB: find books by THEIR POSITION ON A MAP



Embeddings: From Meaning to Math

- With vectors, “similar meaning” is a measurable quantity.

Real systems typically use 1536 dimensions.

"农村在线辅导" → [0.21, -0.83, 0.45, ..., 0.19] (1536 numbers)

"偏远地区教育" → [0.19, -0.81, 0.48, ..., 0.22] ← cosine similarity = 0.92

"披萨外卖" → [-0.55, 0.71, -0.32, ..., 0.81] ← cosine similarity = 0.05

Embedding model = a neural net that maps text / image / audio into these coordinates,
trained so that SIMILAR MEANING → CLOSE DISTANCE.

- OpenAI text-embedding-3-small → 1536 dims
- BGE, M3E, Sentence-BERT → Chinese-friendly OSS alternatives
- CLIP (text+image), Whisper (audio) → same idea, different inputs

Tools: Pinecone (cloud), Weaviate / Milvus (OSS), ChromaDB (local), pgvector (in Postgres)



Databases Summary

- Transaction Processing (OLTP)
 - Relational databases, normalized (2NF-3NF)
- Datawarehousing, structured analytics (OLAP)
 - Relational databases, denormalized
- Fast and flexible data access and manipulation
 - NoSQL databases
- Semantic (meaning-based) search and AI retrieval
 - Vector databases

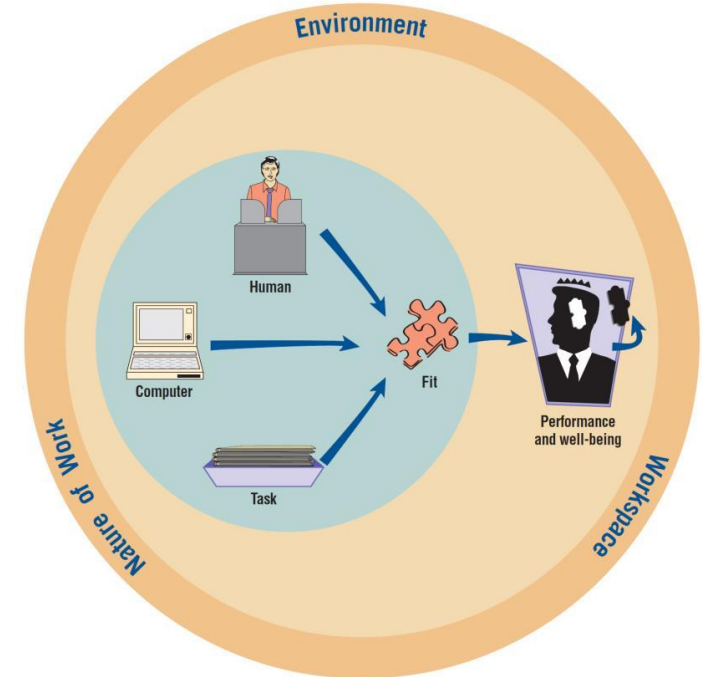


HCI: the Mouth and Ears of the System



HCI: the Pursuit of Fit

- Designing for HCI means
 - ensuring system **functionality** and **usability**,
 - providing effective user **interaction support**,
 - and enhancing a **pleasant user experience**.
- Fit between
 - human
 - the computer
 - the task



Types of User Interfaces

- Natural-language interfaces
- Question-and-answer interfaces
- Menus
- Form-fill interfaces
- Command-language interfaces (CLIs)
- Graphical User Interfaces (GUIs)
- Web interfaces

They are not mutually exclusive. Applications usually combine multiple types, e.g.,
VSCode = GUI + Menus + Command Language + Natural-language (Copilot)。



Types of Feedback

- A button that doesn't light up makes people press it again and again
- Absence of feedback = anxiety = retry = error.

Seven types of feedback every system must provide:

1. Accepting input: "I heard you" (button press, field highlight)
2. Recognizing task: "I understand what you want"
3. Producing more information: "I need more from you"
4. Accepting/processing: "Working on it..." (spinner, progress bar)
5. Rejecting input: "I can't do that" (with WHY)
6. Communicating delay: "This is taking a while" (estimated time)
7. Completing task: "Done. Here's the result."



The AI Era: Interaction as Conversation

- Shift
 - Old: "Teach the user to think like the database."
 - New: "Teach the system to think like the user."
 - Emerging: "Delegate to the system to act on your behalf."

Era	User must...	System provides...
Traditional	Learn SQL, filters, syntax	Exact match on structured data
AI-Native	Speak natural language	Semantic match + generated answer
Agentic	State a GOAL, not a query	Plans and executes multiple steps autonomously



Agentic Interface: the Emerging Form

- The user expresses a GOAL
 - The system PLANS and EXECUTES multiple steps,
 - asking the user only at critical checkpoints.
- Examples in 2026:
 - Cursor / Claude Code / OpenCode: "fix this bug end-to-end"
 - Manus / OpenClaw / Hermes: "book my Shanghai trip"
- It is NOT just "natural language":
 - Natural Language Query: system returns ONE answer; you decide what to do next
 - Agentic: system runs a multi-step PLAN; you supervise, not operate
- What it requires that previous paradigms did not:
 - Tools the agent can call (APIs, code execution, browsing, file edits)
 - Memory across many turns (so the agent does not forget mid-task)
 - Human-in-the-loop checkpoints (approve, interrupt, replay, audit)

For the first time, the user does not OPERATE the system. The user DELEGATES to the system.



An Integrated Case



Reflection

- Databases decide what the system can remember.
- HCI decides what users can ask.
- In our era, the two are converging through AI and vector search
 - users can finally ask anything in natural language,
 - but only if the data has been represented the right way.



Extending the Catalyst (启航杯) Project

- Catalyst Challenge (启航杯)
 - 500+ student business plans (BPs) submitted
 - 20+ judges, each assigned ~30 BPs
 - Prof. Liu opens his email in the morning:
 - 28 PDFs, 5 Word docs, 2 PPTs — all in a ZIP
 - One Excel scoring sheet with 12 dimensions
- His question is NOT: `SELECT * FROM BPs WHERE sector='Education'`
- His question IS: "Which 3 of these are most likely to actually CHANGE the education/entertainment/media industry?"

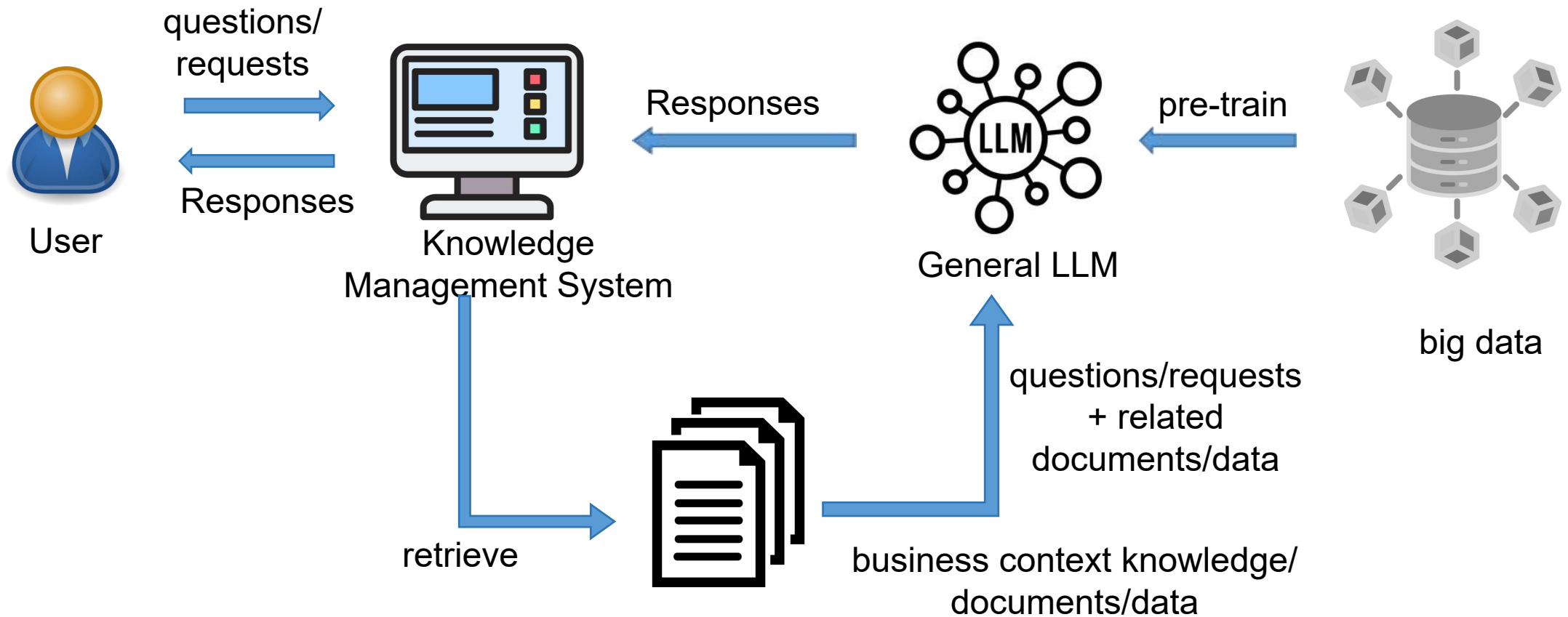


What Database Can Help?

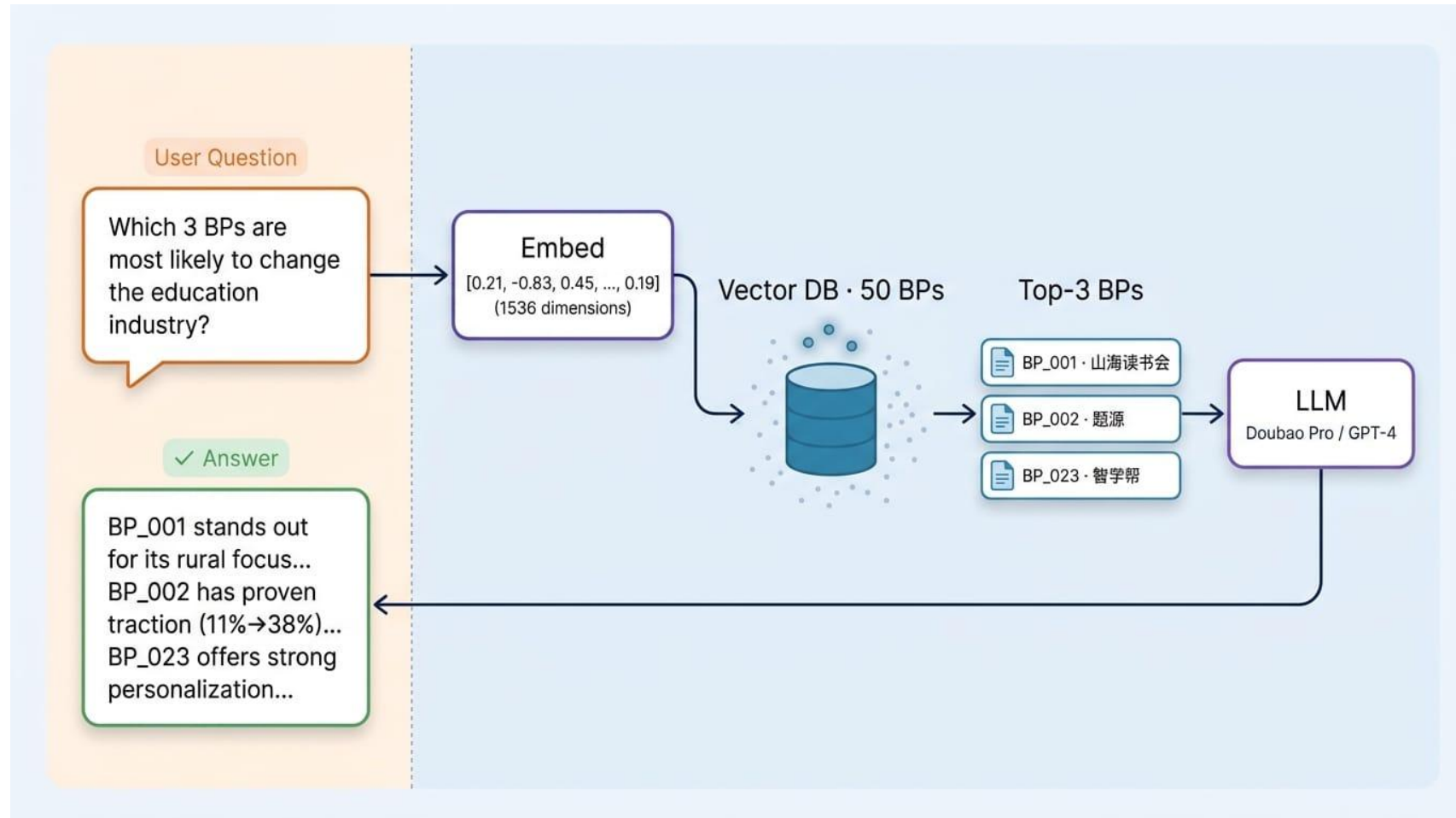
Database	Best query it can do	The result
Relational Databases (e.g., MySQL)	WHERE sector='Education'	Returns ALL education BPs. Cannot rank by "potential to change"
Key-value Databases (e.g., Redis)	GET bp:001 → fetch a single BP by ID	Can only fetch by exact key. No content search at all.
Document Databases (e.g, MongoDB)	find({tags: 'edtech'})	Keyword match, not meaning
Vector Databases	search("change education industry") → top-K BPs by meaning	This IS the answer



Retrival-Augmented Generation (RAG, 检索增强生成)



RAG for the Catalyst Case



Demo for the Catalyst Case



Catalyst 评委助手

RAG 演示 · Lecture 12 · 50 份 BP · text-embedding-3-small + doubao-1-5-pro-32k-250115

embed: online

gen: online

cache: 8

① 预设问题

点击直接运行 (含离线缓存)

1 找出 3 份最有可能改变教育行业的项目, 并说明理由。

已缓存

2 哪些团队在解决碳中和问题? 请列出他们的核心商业模式。

已缓存

3 哪几份 BP 的财务模型最不靠谱? 请说明哪里夸大了。

已缓存

4 找出与去年获奖项目'智慧农场'最相似的 3 份 BP, 并说明相似点与差异点。

已缓存

5 有没有明显抄袭其他公司文案、或自相矛盾的 BP? 请指出具体证据。

已缓存

6 如果我是一个早期阶段的天使投资人, 最想约谈的 3 个团队是哪些? 为什么?

已缓存

7 哪些项目实际上是同一个想法的不同包装? 请按主题聚类。

已缓存

或者输入你自己的问题 (例如: 哪些项目最有可能盈利?)

提问

当前问题

找出 3 份最有可能改变教育行业的项目, 并说明理由。

阶段 1

向量检索

余弦相似度匹配 50 份 BP

1.15s

1 BP 002 题源 教育 相似度 0.449

2 BP 023 智学帮 教育 相似度 0.427

3 BP 027 智学桥梁 教育 相似度 0.426

4 BP 037 智学桥 教育 相似度 0.418

5 BP 016 智学星途 教育 相似度 0.417

阶段 2

拼接 Prompt

检索内容 + 用户问题 → LLM 输入

3852 字符

BP 列表 (50)

按团队/赛道筛选...

001 山海读书会 教育

002 题源 教育

003 盲文之声 教育

004 课灵感 教育

005 碳迹 碳中和

006 光储侠 碳中和

007 二碳 碳中和

008 宇宙速食 消费

009 元梦娃 消费

010 AI牙医 医疗

011 派趣Pizza 消费

012 心语AI AI

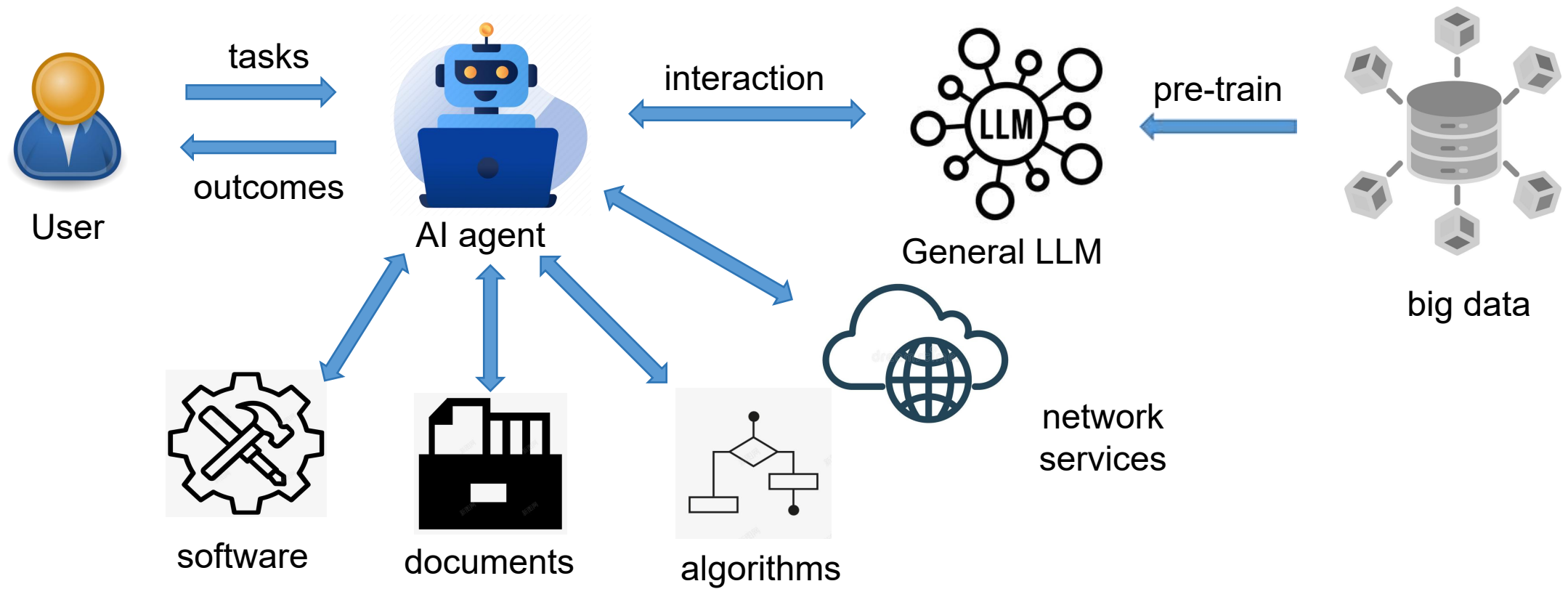
013 稻知道 农业

014 牧云 农业

015 果联 农业

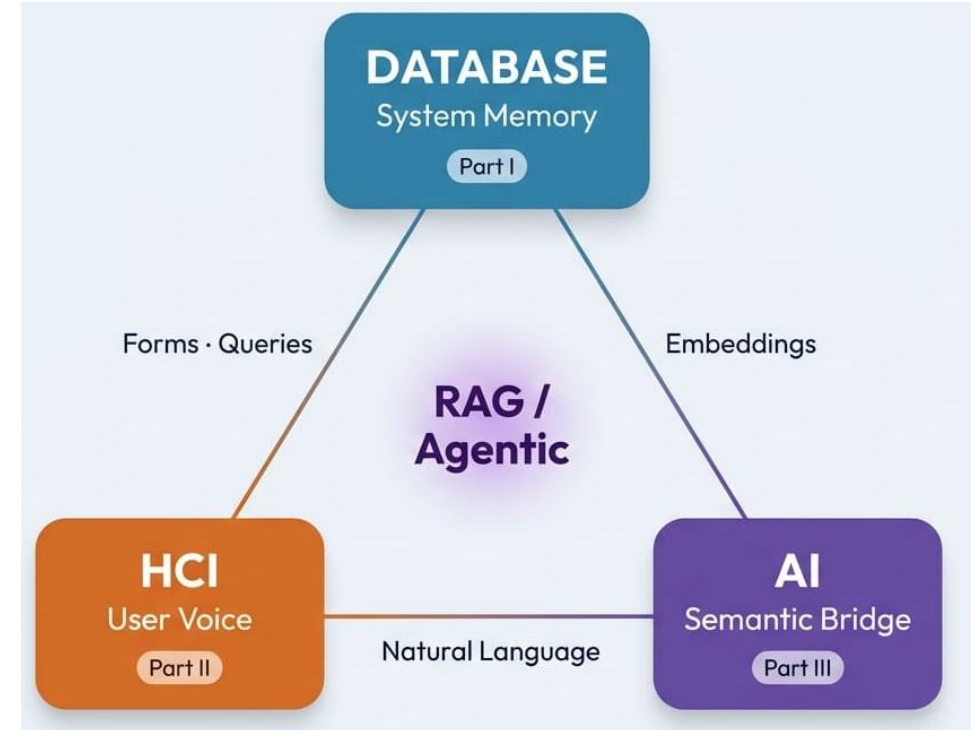


A Further Step: Agentic Interface?



Synthesis: The Database × HCI × AI Triangle

- 1970s–2000s
 - Users learned the database's language (SQL, menus, forms)
- 2010s
 - Mobile/touch made interaction more natural; data stayed structured
- 2020s
 - Vector + LLM let the system understand the USER's language



Tasks

- Individual tasks
 - Reading: Textbook, Chapters 13-14
 - Answer the review questions (on YuKeTang)
 - Answer the additional review question (on WebLearning)
- Group tasks
 - Design your databases and refine your prototype systems.
 - No submission is required for this week.
 - Finally report submission is due **June 12**.
- Class presentations
 - June 1: Group 5 (李稼宜组, 颐和照护) and Group 6 (朱传志组, 极速车队)
 - Content: all your progress with solution design and the prototype system.
 - Time: 10~20 minutes

